WINGS ACADEMY

SYLLABUS FOR WINGS OLYMPIAD EXAMINATION

I. Life Sciences (Biology)

- **Cell Biology (Caskets of Life):**
  identifying parts of a cell (nucleus, cytoplasm, cell wall), and distinguishing between plant and animal cells.
- **Plant Reproduction (Flower to Flower):**
  Parts of a flower (thalamus, calyx, corolla, androecium, gynoecium) and their functions.
  - Process of pollination (self and cross-pollination) and fertilization.
  - Classification of fruits: simple, aggregate, multiple, and false fruits.
- **Health and Nutrition (Food for Health):**
  Nutrients in food (carbohydrates, proteins, fats, vitamins, minerals), their importance, and identifying deficiency diseases.
- **Human Anatomy (For Shape and Strength):**
  The human skeletal system, functions of bones, and types of joints (e.g., hinge joints).
- **Environmental Health (**For a Pollution Free Nature): Causes and effects of air, water, and soil pollution.

II. Physical Sciences (Physics & Chemistry)

- **Energy and its Changes (The Essence of Change):**
  Forms of energy (electrical, light, heat) and energy conversions in daily life.
  - Fossil fuels (LPG, coal) and their dependence on solar energy.
- **Matter and States of Change:**
  Physical vs. chemical changes (permanent vs. temporary).
  - Change of state (melting, evaporation, condensation) due to heat absorption or release.
- **Motion and Force (Along with Motion):**
  Types of motion (rotation, revolution, linear, circular) and force's role in changing a body's state of rest or direction.
  - Speed and velocity concepts.
- **Electricity and Circuits:**
  Function of an Electric Cell, identifying closed vs. open circuits, and distinguishing between **Conductors and Insulators**.
- **Magnetism (Attraction and Repulsion):**
  Properties of magnets, magnetic poles, and the concept of a magnetic field.
  - Classifying magnetic and non-magnetic substances.
- **Chemistry of Mixtures (Mix and Separate):**
  Methods for separating mixtures and understanding pure substances versus mixture

General aptitude

1. Logical & Abstract Reasoning

- **Series Completion:** Finding the next item in a sequence of numbers, letters, or complex shapes

  (e.g 1,4,9,16, _?).

- **Coding-Decoding:** If "APPLE" is written as "BQQMF," how is "MANGO" written?
- **Blood Relations:** (e.g., "The man in the photo is my mother's only son. Who is he?").
- **Mirror & Water Images:** Visualizing how a shape or word looks when reflected.

2. Quantitative & Arithmetical Aptitude

- **Clocks & Calendars:** Calculating the angle between clock hands or finding the day of the week for a future date.
- **Direction Sense:** A person walks 5km North, turns right, walks 3km... where are they now?

3. Verbal Aptitude (Critical Thinking)

- **Analogies:** Identifying relationships (e.g., *Doctor : medicine :: postman:* _____ *?*).
- **Odd One Out:** Finding the "intruder" in a group based on hidden properties

  (e.g., Nitrogen, Oxygen, Carbon Dioxide, Water

  Odd One: Water (It is a liquid at room temperature;)